

Efficacy and safety of ecopipam for Tourette syndrome: An updated systematic review and meta-analysis of randomized controlled trials

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Abstract

Background: Tourette syndrome is a neurodevelopmental disorder characterized by persistent motor and vocal tics. Dopamine D2 receptor antagonists can reduce tic severity but are frequently limited by metabolic, sedative, and extrapyramidal adverse effects. Ecopipam is a selective dopamine D1/D5 receptor antagonist with a potentially distinct tolerability profile. **Objectives:** To evaluate the efficacy and safety of ecopipam in patients with Tourette syndrome using updated randomized evidence. **Design:** Systematic review and random-effects meta-analysis of randomized placebo-controlled trials. **Data sources and methods:** PubMed/MEDLINE, Embase, Scopus, [ClinicalTrials.gov](https://clinicaltrials.gov), reference lists, and conference sources were searched from database inception to May 31, 2026. The primary outcome was change in Yale Global Tic Severity Scale Total Tic Score. Headache and insomnia were evaluated as safety outcomes. Treatment estimates were pooled using random-effects models with Hartung–Knapp adjustment. Risk of bias was assessed using the Cochrane RoB 2 tool. **Results:** Three randomized trials involving 297 unique randomized participants were included. Ecopipam improved tic severity compared with placebo (mean difference, -3.72 ; 95% confidence interval, -5.96 to -1.47). The result remained significant after exclusion of the phase 3 randomized-withdrawal trial (mean difference, -3.33 ; 95% confidence interval, -4.85 to -1.81). No statistical heterogeneity was detected for efficacy outcomes, although the small number of trials limits interpretation. Headache was not significantly increased (risk ratio, 1.36; 95% confidence interval, 0.52–3.51). Insomnia showed an imprecise, nonsignificant association (risk ratio, 2.44; 95% confidence interval, 0.07–79.93; $I^2=62.2\%$). **Conclusion:** Short-term randomized evidence suggests that ecopipam reduces tic severity. Evidence concerning insomnia and long-term psychiatric and metabolic safety remains insufficient, requiring larger independent trials with longer follow-up.

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Plain language summary

Tourette syndrome is a condition that causes repeated movements and sounds called tics. Medicines currently used to treat tics can be effective, but they may also cause unwanted side effects such as weight gain, sleepiness, and movement problems. Ecopipam is a new medicine that works differently from currently available treatments. We reviewed and combined the results of three clinical trials involving 297 people with Tourette syndrome to better understand how well ecopipam works and how safe it is. Our analysis showed that people taking ecopipam experienced greater improvements in tic severity than those taking placebo (an inactive treatment). The benefits were consistent across studies. We also found that common side effects such as headache and insomnia were not clearly more frequent with ecopipam than with placebo, although the available safety data were limited. Overall, the current evidence suggests that ecopipam may be an effective and generally well-tolerated treatment for Tourette syndrome. However, additional large studies with longer follow-up are needed to confirm its long-term benefits and safety.

Keywords

ecopipam, Tourette syndrome, tic disorder, dopamine D1 receptor antagonist, YGTSS, meta-analysis, randomized controlled trial

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Introduction

Tourette syndrome is defined by multiple motor tics and at least one vocal tic persisting for more than 1 year, with onset before adulthood.¹ Tic expression varies across development and may fluctuate with stress, arousal, fatigue, and contextual demands. Although many patients improve during adolescence, a clinically important subgroup has persistent tics associated with pain, stigma, social avoidance, sleep disruption, academic impairment, and reduced quality of life.^{2,3} A population-based systematic review has also characterized the prevalence of Tourette syndrome across studied populations.⁴ The Yale Global Tic Severity Scale (YGTSS) remains the principal clinician-rated endpoint in therapeutic trials because it captures motor and vocal tic number, frequency, intensity, complexity, and interference.⁵

Current treatment is individualized. Behavioural therapy, particularly comprehensive behavioural intervention for tics, is recommended when available and acceptable.^{6,7} Pharmacological options include alpha-2 adrenergic agonists, antipsychotics, and vesicular monoamine transporter 2 inhibitors, but no medication is ideal for all patients.^{8,9} Dopamine D2 receptor-blocking agents can reduce tics, yet their use is often limited by somnolence, akathisia, parkinsonism, weight gain, metabolic effects, prolactin elevation, and concerns about tardive syndromes during longer treatment.^{10,11} These limitations support continued evaluation of targeted dopaminergic strategies that may preserve anti-tic efficacy while improving tolerability.

Ecopipam is an orally administered selective dopamine D1/D5 receptor antagonist. The therapeutic rationale is based on the role of cortico-striato-thalamo-cortical circuits and dopaminergic signalling in tic generation.^{12,13} An early open-label adult study suggested tic reduction and acceptable short-term tolerability, providing a basis for randomized paediatric evaluation.¹⁴ Subsequent randomized trials included a placebo-controlled crossover trial, a larger 12-week parallel-group trial, and a phase 3 randomized-withdrawal study evaluating maintenance of effect.^{15–17} Because randomized evidence has recently expanded, a focused synthesis is needed for clinicians and investigators. We therefore performed an updated systematic review and meta-analysis of randomized controlled trials of ecopipam for Tourette syndrome, emphasizing efficacy, headache, insomnia, and robustness of the primary outcome.

Two previous systematic reviews evaluated ecopipam for Tourette syndrome. Panda et al. reported a significant reduction in YGTSS-TTS based predominantly on the two earlier randomized trials, whereas Darwish et al. reported a nonsignificant week-4 difference in YGTSS-TTS but improvements in selected global clinical measures. Both reviews were based on searches conducted before publication of the 2026 phase 3 randomized-withdrawal trial. An updated synthesis restricted to randomized placebo-controlled evidence and incorporating the phase 3 data was therefore warranted.^{18,19}

Methods

Protocol and reporting standards

This systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement and recommendations from the Cochrane Handbook for Systematic Reviews

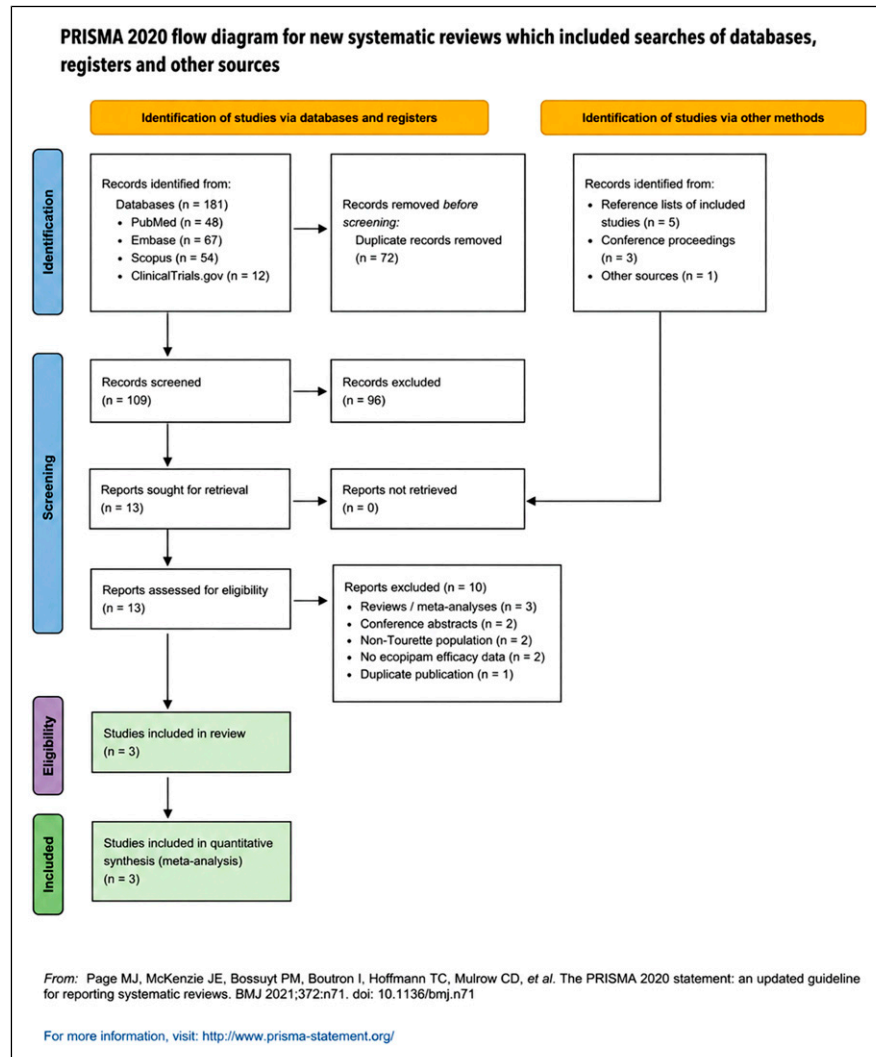


Figure 1. PRISMA 2020 flow diagram showing study identification, screening, eligibility assessment, and inclusion for the updated systematic review and meta-analysis of ecopipam in Tourette syndrome.

of Interventions^{20,21} (Figure 1). The review question was developed using the Population, Intervention, Comparator, Outcomes, and Study Design (PICOS) framework. Eligible participants were children, adolescents, or adults with Tourette syndrome; the intervention was ecopipam; the comparator was placebo; and only randomized controlled trials (RCTs) were included.

Literature search strategy

A comprehensive literature search was conducted in PubMed/MEDLINE, Embase, Scopus, and [ClinicalTrials.gov](https://www.clinicaltrials.gov/) from database inception to **31 May 2026**, without restrictions on publication date. Additional potentially eligible studies were identified through manual screening of the reference lists of included articles, relevant systematic reviews, conference proceedings, and clinical trial records. The search terms included combinations of “Tourette syndrome,” “Tourette disorder,” “tic disorder,” “ecopipam,” “SCH39166,” “dopamine D1 receptor antagonist,” “randomized,” “placebo,” and “clinical trial.” The search strategy was adapted for each database. The complete database-specific search strategies, including the date searched and the number of records retrieved, are provided in [Supplementary Table S1](#).

Study selection

Two reviewers independently screened titles and abstracts, followed by full-text assessment of potentially eligible articles. Disagreements were resolved through discussion and consensus.

Studies were eligible if they: (1) enrolled participants with Tourette syndrome; (2) used a randomized controlled design; (3) compared ecopipam with placebo; and (4) reported extractable efficacy or safety data. Observational studies, case reports, reviews, meta-analyses, editorials, conference abstracts without sufficient outcome data, animal studies, and duplicate publications were excluded. When multiple reports described the same study population, the most complete peer-reviewed publication was retained.

Data extraction

Two investigators independently extracted data using a standardized form. Extracted variables included study characteristics, publication year, sample size, participant demographics, baseline tic severity, treatment duration, intervention details, efficacy outcomes, and adverse events. Any discrepancies were resolved through review of the original reports and consensus discussion.

Outcomes

The primary efficacy outcome was change in Yale Global Tic Severity Scale Total Tic Score (YGTSS-TTS) from baseline to the randomized study endpoint. The YGTSS-TTS is a validated clinician-rated measure assessing tic number, frequency, intensity, complexity, and interference, with higher scores indicating greater tic severity.

Additional efficacy measures, including the Yale Global Tic Severity Scale Global Severity Score (YGTSS-GSS), Clinical Global Impression–Improvement (CGI-I), and Clinical Global Impression–Severity (CGI-S), were extracted and summarized descriptively when available. For safety analyses, headache and insomnia were selected as prespecified outcomes because they were clinically relevant and consistently reported across the included trials.

Statistical analysis

Meta-analyses were performed using R software (R Foundation for Statistical Computing, Vienna, Austria) with the *meta* and *metafor* packages.^{22,23} Continuous outcomes were pooled using the Generic Inverse Variance (GIV) method to calculate pooled mean differences (MDs) with 95% confidence intervals (CIs). Individual studies were weighted according to the inverse of the variance of their effect estimates. When standard errors were not directly reported, they were calculated from the published 95% confidence intervals using established statistical methods. Dichotomous outcomes were pooled as risk ratios (RRs) with 95% CIs.

Random-effects models were applied for all analyses because of anticipated methodological and clinical heterogeneity arising from differences in trial design, treatment duration, and participant characteristics. Hartung–Knapp adjustment was used to provide more reliable confidence interval estimation given the small number of included studies.

Statistical heterogeneity was assessed using Cochran's Q test, the I^2 statistic, and between-study variance (τ^2). Negative MD values favored ecopipam for efficacy outcomes. A prespecified sensitivity analysis excluding the phase 3 randomized-withdrawal trial was performed to evaluate the robustness of the primary efficacy findings and determine whether the pooled estimate was influenced by its enrichment design.

Risk of bias assessment

Risk of bias was independently assessed by two reviewers using the Cochrane Risk of Bias 2 (RoB 2) tool.²⁴ Domains evaluated included bias arising from the randomization process, deviations from intended interventions, missing outcome data, outcome measurement, and selection of reported results. Each study was categorized as having low risk of bias, some concerns, or high risk of bias according to RoB 2 guidance (Figure 2).

Because fewer than ten studies were included, formal assessment of publication bias using funnel plots or statistical tests was not performed. The certainty of evidence was interpreted narratively considering study limitations, consistency of findings, precision of estimates, and the size of the available evidence base.



Figure 2. Risk of bias assessment using the Cochrane RoB 2 tool across randomization, deviations from intended interventions, missing outcome data, outcome measurement, selective reporting, and overall bias.

Results

Study selection

The literature search identified 181 records from electronic databases and clinical trial registries, including PubMed ($n = 48$), Embase ($n = 67$), Scopus ($n = 54$), and [ClinicalTrials.gov](https://www.clinicaltrials.gov) ($n = 12$). An additional nine records were identified through other sources, including reference list screening, conference proceedings, and manual searches. After removal of 72 duplicate records, 118 unique records remained for title and abstract screening. Of these, 96 records were excluded because they were reviews, editorials, non-randomized studies, animal studies, or were unrelated to Tourette syndrome or ecopipam therapy. Thirteen reports underwent full-text assessment for eligibility. Following detailed review, ten reports were excluded for predefined reasons, including review articles or meta-analyses ($n = 3$), conference abstracts without usable outcome data ($n = 2$), studies involving non-Tourette populations ($n = 2$), absence of extractable efficacy outcomes ($n = 2$), and duplicate publication of previously reported data ($n = 1$). Ultimately, three randomized controlled trials met all eligibility criteria and were included in both the qualitative synthesis and quantitative meta-analysis (Figure 1).

Study characteristics

The three included randomized controlled trials enrolled a total of 297 participants with Tourette syndrome. Sample sizes ranged from 40 to 153 participants. Two studies were conducted exclusively in children and adolescents, whereas the phase 3 trial included a broader paediatric population with moderate-to-severe tic symptoms. Study duration varied from 4 to 24 weeks.

Gilbert et al. conducted the first randomized double-blind crossover study evaluating ecopipam in paediatric Tourette syndrome. Participants received ecopipam and placebo in a crossover fashion over four weeks, with tic severity assessed using the Yale Global Tic Severity Scale Total Tic Score (YGTSS-TTS). The study demonstrated significant short-term reductions in tic severity and provided preliminary evidence supporting D1 receptor antagonism as a therapeutic strategy for Tourette syndrome.¹⁵

The subsequent D1AMOND phase 2b trial was a multicentre, randomized, placebo-controlled study that evaluated ecopipam over a 12-week treatment period. Participants receiving ecopipam achieved greater reductions in YGTSS-TTS compared with placebo and showed improvement across several secondary efficacy measures, including clinician-rated global improvement scores.¹⁶

The most recent phase 3 trial employed an enrichment and randomized-withdrawal design. Participants initially received open-label ecopipam, after which responders were randomized to continue ecopipam or switch to placebo. This study demonstrated maintenance of tic improvement among participants who continued ecopipam treatment while providing additional long-term safety data.¹⁷

Across all studies, participants had moderate-to-severe baseline tic severity, with mean YGTSS-TTS scores generally ranging from approximately 28 to 35 points. The YGTSS-TTS served as the primary efficacy endpoint in the earlier trials and as an exploratory efficacy endpoint during the randomized-withdrawal phase of the phase 3 trial. Clinical Global Impression scales²⁵ and adverse-event monitoring^{15–17} were also used to assess overall clinical benefit and treatment tolerability. Detailed baseline characteristics are presented in Table 1.

Table 1. Baseline characteristics of included randomized controlled trials.

Study	Design and duration	Population	Randomized sample	Baseline severity	Key notes
Gilbert et al. 2018 ¹⁵	Randomized, double-blind, placebo-controlled crossover; 4 weeks per period	Children and adolescents with Tourette syndrome	n=40	Moderate to severe tic burden by YGTSS-TTS	Short-term paediatric proof-of-concept trial.
Gilbert et al. 2023 ¹⁶	Randomized, double-blind, parallel-group; 12 weeks	Children and adolescents with moderate to severe Tourette syndrome	n=153	Mean YGTSS-TTS in moderate to severe range	Largest initiation trial; primary endpoint was YGTSS-TTS change.
Gilbert et al. 2026 ¹⁷	Phase 3 randomized-withdrawal trial; up to 24 weeks	Children, adolescents, and adults with a response during the open-label ecopipam phase	n=104 randomized in withdrawal phase	Responder-enriched population	Responder-enriched randomized-withdrawal trial designed to assess maintenance of effect; the primary analysis evaluated time to relapse in pediatric participants, with the adult analysis exploratory.

Primary efficacy outcome

All three randomized trials contributed data to the primary efficacy analysis evaluating change in YGTSS-TTS. Pooled analysis demonstrated that ecopipam produced a statistically significant improvement in tic severity compared with placebo. The overall mean difference was -3.72 points (95% CI, -5.96 to -1.47), favoring ecopipam. No statistical heterogeneity was detected; however, the small number of included trials limits the precision and interpretability of heterogeneity estimates (Figure 3).

The YGTSS-TTS evaluates tic number, frequency, intensity, complexity, and interference, although the clinical importance of a pooled mean difference should be interpreted alongside responder and global-improvement outcomes.^{5,26} No included trial demonstrated a treatment effect favoring placebo, and the direction of effect was consistent across the randomized studies.^{15–17}

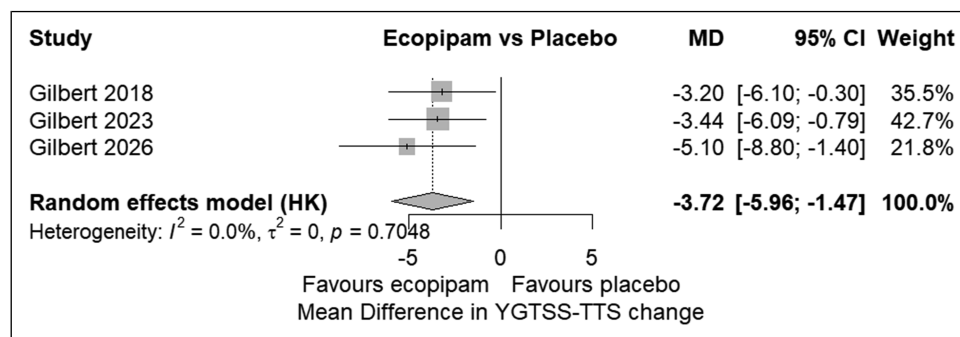


Figure 3. Forest plot for the primary efficacy outcome: change in YGTSS-TTS comparing ecopipam versus placebo. Negative mean differences favour ecopipam.

Sensitivity analysis

To evaluate the robustness of the primary efficacy findings, a prespecified sensitivity analysis was performed excluding the phase 3 randomized-withdrawal trial, which differed methodologically from the earlier parallel-group studies due to its enrichment design. The pooled estimate remained statistically significant, with a mean difference of -3.33 points (95% CI, -4.85 to -1.81). Again, no statistical heterogeneity was detected; however, the small number of included trials limits the precision and interpretability of heterogeneity estimates.

Notably, the magnitude of benefit observed in the sensitivity analysis was highly similar to that observed in the primary analysis, suggesting that the overall efficacy estimate was not disproportionately influenced by the enriched responder population included in the phase 3 study. These findings provide additional reassurance regarding the robustness and generalizability of the treatment effect observed with ecopipam (Figure 4).

Safety outcomes

Safety analyses focused on the most consistently reported adverse events across studies. Overall adverse event rates were generally mild to moderate in severity, and treatment discontinuation due to adverse events was uncommon.

Headache

Headache was reported in all three randomized trials and was therefore included in pooled analysis. Meta-analysis demonstrated no statistically significant difference between ecopipam and placebo (RR, 1.36; 95% CI, 0.52–3.51). No statistical heterogeneity was detected for headache ($I^2=0\%$); however, because only three trials contributed to this analysis, the heterogeneity estimate has limited precision and should be interpreted cautiously (Figure 5). Although headache occurred numerically more frequently among ecopipam-treated participants, the confidence interval crossed unity and did not support a definitive association between treatment and increased headache risk.

Insomnia

Insomnia was reported less consistently across studies but was considered clinically relevant given the neuropsychiatric mechanism of ecopipam. The pooled analysis demonstrated a numerically increased risk of insomnia among ecopipam-treated participants (RR, 2.44; 95% CI, 0.07–79.93). However, the confidence interval was extremely wide and crossed the null value, reflecting substantial statistical uncertainty. Moderate heterogeneity was observed ($I^2 = 62.2\%$), suggesting variation in insomnia reporting between studies (Figure 6). Consequently, no definitive conclusion regarding insomnia risk can be drawn from the currently available evidence. A summary of the pooled efficacy and safety findings is presented in Table 2.

Risk of bias assessment

Risk of bias was assessed using the Cochrane Risk of Bias 2 (RoB 2) tool. Overall methodological quality was favorable. All included studies employed randomized allocation procedures and double-blind treatment administration. Outcome assessment relied on validated clinician-rated scales, reducing the likelihood of measurement bias.

The phase 2b and phase 3 studies were judged to have low risk of bias across most domains. Some concerns were identified in the crossover trial because of potential carryover effects and complexities associated with crossover study design.

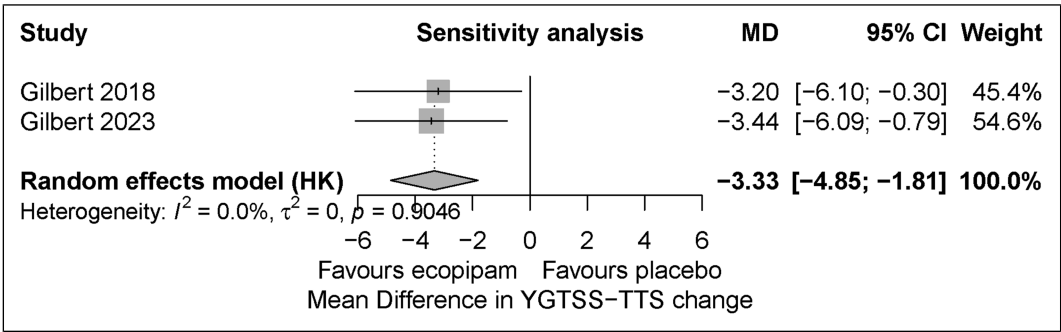


Figure 4. Sensitivity analysis excluding the phase 3 randomized-withdrawal trial. Negative mean differences favour ecopipam.

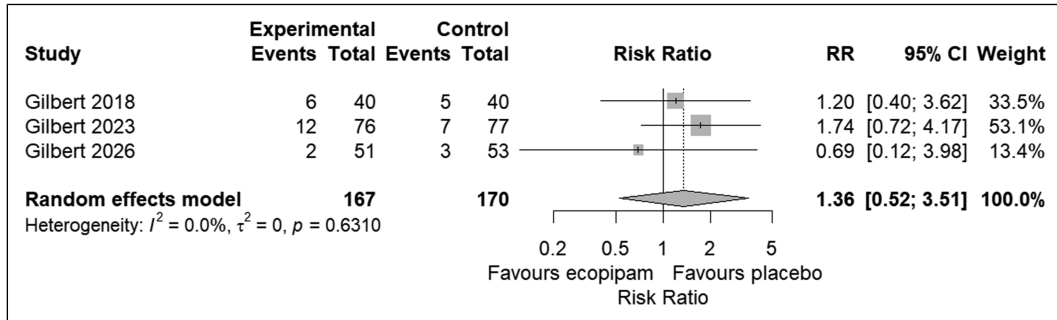


Figure 5. Forest plot for headache. Risk ratios greater than 1 indicate higher event frequency with ecopipam.

Additional concerns arose from attrition and enrichment procedures used in the randomized-withdrawal study, which may potentially overestimate maintenance benefits among initial responders. Nevertheless, outcome reporting was consistent with prespecified endpoints, and selective reporting bias was considered unlikely.

Overall, the certainty of the randomized evidence was strengthened by low heterogeneity, consistency of treatment effects, rigorous blinding procedures, and use of standardized efficacy measures. The detailed risk-of-bias assessment is presented in [Figure 2](#).

Discussion

This updated systematic review and meta-analysis synthesizes the currently available randomized evidence evaluating ecopipam for Tourette syndrome. Across three randomized controlled trials involving 297 participants, ecopipam produced a statistically significant reduction in tic severity compared with placebo, with consistent findings across studies and no observed heterogeneity; however, the small number of included trials limits the precision and interpretability of heterogeneity estimates. Importantly, the treatment effect remained significant after exclusion of the phase 3 randomized-withdrawal trial, suggesting that the observed benefit was not solely driven by enrichment of prior responders. Taken together, these findings support ecopipam as a promising therapeutic option for tic reduction and provide further evidence that selective dopamine D1 receptor antagonism may represent a viable treatment strategy in Tourette syndrome.^{12–17}

The present findings should be considered in relation to previous evidence syntheses. Panda et al. reported a pooled YGTSS-TTS benefit of approximately 3 points based mainly on the two earlier randomized trials, whereas Darwish et al. reported a nonsignificant week-4 YGTSS-TTS difference. The current analysis differs by restricting inclusion to randomized placebo-controlled trials, incorporating the recently published phase 3 randomized-withdrawal study, using endpoint-level treatment estimates, and performing a sensitivity analysis excluding the responder-enriched withdrawal trial. Differences among pooled estimates may therefore reflect study eligibility, endpoint timing, and analytical approach rather than fundamentally conflicting evidence.^{18,19}

Current pharmacologic management of Tourette syndrome is dominated by dopamine D2 receptor antagonists, including haloperidol, pimozide, aripiprazole, and risperidone.^{8,9,27} Although these medications can effectively reduce tic severity, their clinical utility is frequently limited by adverse effects such as weight gain, metabolic disturbances, sedation, hyperprolactinemia, and extrapyramidal symptoms.^{8–11} Consequently, there remains substantial interest in developing therapies that maintain efficacy while improving tolerability. Ecopipam differs fundamentally from existing anti-tic

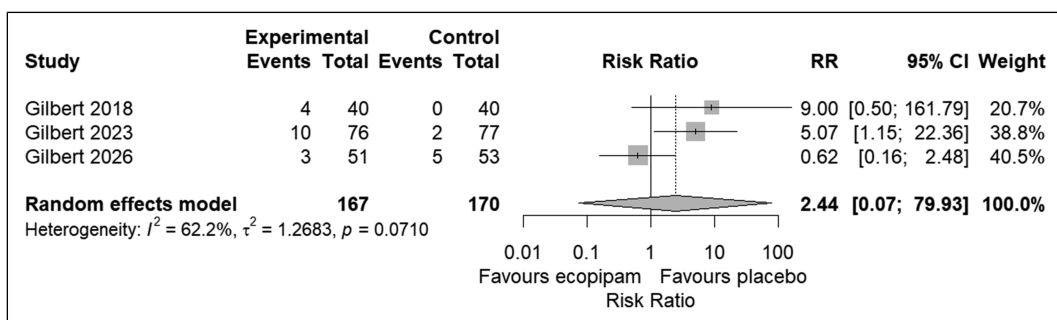


Figure 6. Forest plot for insomnia. Risk ratios greater than 1 indicate higher event frequency with ecopipam.

Table 2. Summary of pooled efficacy and safety outcomes.

Outcome	Studies (n)	Participants (n)	Effect measure	Pooled effect (95% CI)	P value	I^2 (%)
Primary efficacy outcome						
Change in YGTSS-TTS	3	297	MD	-3.72 (-5.96 to -1.47)	0.013	0
Sensitivity analysis						
Change in YGTSS-TTS (excluding phase 3 randomized-withdrawal trial)	2	193	MD	-3.33 (-4.85 to -1.81)	0.006	0
Safety outcomes						
Headache	3	297	RR	1.36 (0.52 to 3.51)	0.53	0
Insomnia	3	297	RR	2.44 (0.07 to 79.93)	0.62	62.2

Abbreviations: CI, confidence interval; I^2 , inconsistency statistic; MD, mean difference; RR, risk ratio; YGTSS-TTS, Yale Global Tic Severity Scale Total Tic Score. Continuous outcomes were pooled using the Generic Inverse Variance method under random-effects models with Hartung-Knapp adjustment. Dichotomous outcomes were pooled as risk ratios using random-effects models. When standard errors were not directly reported, they were calculated from published 95% confidence intervals.

medications because it selectively antagonizes dopamine D1/D5 receptors without directly blocking D2 receptors. Experimental and neurobiological evidence suggests that both D1- and D2-mediated signaling pathways contribute to dysfunction within cortico-striato-thalamo-cortical circuits implicated in Tourette syndrome.^{4,12} Therefore, modulation of D1 receptor activity may provide therapeutic benefit while potentially avoiding some of the adverse effects associated with chronic D2 receptor blockade.

The magnitude of improvement observed in the present analysis warrants consideration. The pooled mean difference of approximately 3.7 points on the YGTSS-TTS may appear modest numerically; however, interpretation of changes in tic severity scales is complex. The clinical significance of a given reduction depends not only on the absolute score change but also on baseline tic burden, functional impairment, psychosocial impact, and patient perception of improvement.^{26,28} In clinical practice, even moderate reductions in tic frequency or intensity can translate into meaningful improvements in academic performance, social interactions, self-esteem, and quality of life. Furthermore, several included trials reported favorable findings on global improvement measures, suggesting that the observed reductions in YGTSS-TTS were accompanied by clinically perceptible benefits.^{15–17} The absence of heterogeneity across studies; however, the small number of included trials limits the precision and interpretability of heterogeneity estimates.

An important aspect of the current findings is the consistency between early proof-of-concept studies and more recent phase 3 investigations. Historically, many candidate therapies for Tourette syndrome have demonstrated encouraging results in preliminary studies but failed to confirm efficacy in larger trials. In contrast, ecopipam has shown relatively stable treatment effects across multiple randomized studies spanning different phases of clinical development.^{15–17} The persistence of benefit in sensitivity analyses suggests that efficacy is not restricted to a specific trial design and supports the overall robustness of the evidence base. Nevertheless, the total number of participants remains limited compared with established pharmacologic interventions, highlighting the need for additional confirmatory research.

Safety considerations are particularly important when evaluating therapies for Tourette syndrome because many patients require long-term treatment beginning in childhood or adolescence. In the present analysis, headache was not significantly more frequent with ecopipam than placebo, and no consistent signal of increased risk emerged across studies. Insomnia requires particular consideration. In the phase 2b parallel-group trial, insomnia occurred in 10 of 76 participants receiving ecopipam compared with 2 of 77 receiving placebo, representing a notable trial-level imbalance.¹⁶ However, the direction and magnitude of the association were not consistent across the other randomized studies, including the phase 3 randomized-withdrawal period. Consequently, although the pooled association was not statistically significant, its extremely wide confidence interval and moderate heterogeneity indicate substantial uncertainty rather than evidence of no risk. Sleep-related adverse events should therefore be prospectively defined, systematically collected, and monitored in larger and longer-duration trials. Importantly, none of the included trials identified a clear signal for extrapyramidal symptoms, tardive dyskinesia, or substantial metabolic complications, adverse effects that often influence treatment selection when D2 receptor antagonists are prescribed.^{8–11,27} Although indirect comparisons should be interpreted cautiously, the available evidence raises the possibility that ecopipam may offer a more favorable tolerability profile than traditional antipsychotic therapies.

Behavioral therapy also has established efficacy and remains an important nonpharmacological treatment and comparator for future ecopipam trials.²⁹

Another consideration is the high prevalence of psychiatric comorbidities among individuals with Tourette syndrome. Attention-deficit/hyperactivity disorder, obsessive-compulsive disorder, anxiety disorders, depressive symptoms, and disruptive behavioral disorders may contribute substantially to functional impairment.^{30,31} Stress may also modulate tic expression and the clinical course of Tourette syndrome.³² These comorbid conditions may influence both treatment response and adverse event reporting. Because the available randomized trials were not powered to examine outcomes according to psychiatric comorbidity status, it remains unclear whether ecopipam provides differential benefits in specific patient subgroups. Future studies should incorporate stratified analyses and standardized psychiatric assessments to better characterize the relationship between tic improvement and broader neuropsychiatric outcomes.

This review has several strengths. It represents the most up-to-date quantitative synthesis of randomized evidence evaluating ecopipam in Tourette syndrome and incorporates recently published phase 3 data. Methodological rigor was enhanced through adherence to PRISMA recommendations, duplicate screening and data extraction, formal risk-of-bias assessment using RoB 2, and sensitivity analyses designed to evaluate the robustness of the primary findings. The absence of statistical heterogeneity across efficacy analyses further supports the reliability of the pooled estimates; however, the small number of included trials limits the precision and interpretability of heterogeneity estimates.

The current findings have important implications for future research and clinical practice. If ongoing development confirms long-term efficacy and safety, ecopipam could expand the therapeutic landscape for Tourette syndrome by providing a non-D2 dopaminergic treatment option. Future trials should incorporate longer follow-up periods, standardized reporting of YGTSS outcomes, comprehensive psychiatric safety assessments, patient-reported outcome measures, and direct comparisons with established therapies such as aripiprazole, risperidone, or comprehensive behavioral intervention for tics (CBIT). Such studies will be essential for determining the optimal place of ecopipam within treatment algorithms and for identifying patient populations most likely to benefit from therapy.

Limitations

Several limitations should also be acknowledged. First, only three randomized trials were available, resulting in a relatively small cumulative sample size. Second, trial designs differed substantially, including crossover, parallel-group, and randomized-withdrawal methodologies. Although sensitivity analyses yielded similar results, design-related differences may still influence pooled estimates. Third, aggregate published data prevented detailed subgroup analyses according to age, sex, baseline tic severity, psychiatric comorbidities, or concomitant medication use. Fourth, follow-up durations were relatively short, limiting assessment of long-term efficacy and uncommon adverse events. Finally, because ecopipam remains an investigational therapy, independent replication outside sponsor-supported clinical development programs remains limited.

Conclusion

In this updated meta-analysis of randomized controlled trials, ecopipam significantly reduced tic severity compared with placebo in patients with Tourette syndrome. The treatment effect remained robust in sensitivity analyses and was not accompanied by a significant increase in headache, while evidence regarding insomnia remained inconclusive. Current evidence supports ecopipam as a promising selective dopamine D1/D5 receptor antagonist with the potential to address an important unmet need in Tourette syndrome management. Nevertheless, larger independent studies with longer follow-up are required to establish long-term safety, durability of response, and comparative effectiveness relative to existing treatment options.

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Ethical considerations

Not applicable. This systematic review and meta-analysis used data from previously published studies and did not involve direct recruitment of participants or access to identifiable individual-level data. Institutional ethics approval and participant consent were therefore not required.

Author contributions

Aasim Ali: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Software, Visualization, Project administration, and Writing – original draft. Abdullah Shahid: Conceptualization, Methodology, Validation, Supervision, and Writing – review and editing. Muhammad Usman: Methodology, Investigation, Data curation, Formal analysis, Visualization, and Writing – review and editing. Asad Ullah Ansari: Investigation, Data curation, Validation, and Writing – review and editing. Labeeba Abdul Ghafoor: Investigation, Data curation, Validation, and Writing – review and editing. Csilla Dézsi: Methodology, Validation, and Writing – review and editing. Viktor Gulyás-Oldal: Methodology, Validation, and Writing – review and editing. Károly Szili: Supervision, Validation, Investigation, and Writing – review and editing. Muhammad Abdullah Ali: Investigation, Data curation, Validation, and Writing – review and editing. Usama Saleem: Investigation, Data curation, Validation, and Writing – review and editing. Muhammad Asad Shabbir: Investigation, Data curation, Validation, and Writing – review and editing. Armaghana Abdullah: Investigation, Data curation, Validation, and Writing – review and editing. Mukesh Kumar Sharma: Supervision, Project administration, Methodology, Validation, and Writing – review and editing. Muhammad Talha: Investigation, Data curation, Validation, and Writing – review and editing. All authors critically reviewed the manuscript, approved the final version for publication, and agreed to be accountable for all aspects of the work.

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Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

All data analyzed in this systematic review and meta-analysis were obtained from the published studies and publicly available trial records cited in the manuscript. The extracted dataset and analytical materials are available from the corresponding author on reasonable request.

Trial registration

Not registered.

AI statement

An AI-assisted language tool was used solely to improve grammar, clarity, and formatting. No AI tool was used to generate study data or independently conduct or interpret the statistical analyses. All text and cited sources were reviewed and verified by the authors, who take full responsibility for the manuscript.

Supplemental material

Supplemental material for this article is available online.

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